

Electrifying Your Fleet: Turning a Cost Center into a Revenue Producer with V2X

Meta Description: CFOs in organizations looking into fleet electrification have an opportunity to turn a traditional cost center into a revenue generator. A V2X strategy is the key.

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Regulators and government organizations at every level are aggressively pursuing decarbonization solutions, with beneficial electrification receiving significant attention. Mitigating human-caused climate change and advancing sustainability practices are two of the primary goals of the efforts. And for CFOs of organizations considering fleet electrification, there's an opportunity to turn a cost center into a revenue generator with a V2X solution.

Beneficial and Fleet Electrification Fit Together Beautifully

So, what is beneficial electrification (BE), and how does fleet electrification contribute to the dual goals of climate change mitigation and sustainability? The Environmental and Energy Study Institute (EESI) [defines beneficial electrification](#) as follows:

***“Beneficial electrification** (or strategic electrification) is a term for replacing direct fossil fuel use (e.g., propane, heating oil, gasoline) with electricity in a way that reduces overall emissions and energy costs. This can include switching to an electric vehicle – as long as the end-user and the environment both benefit.”*

According to the [Environmental Protection Agency](#) (EPA), in 2021, the transportation sector was responsible for 29% of all greenhouse gas (GHG) emissions. The transportation sector includes aircraft, trains, cars, and trucks. Light-duty vehicles of all types account for 58% of the transportation sector's GHG emissions.

The high percentage of GHG emissions from light-duty vehicles makes them highly desirable targets as a tangible demonstration of decarbonization. While per-vehicle contributions are small, the impacts are significant when multiplied by tens of thousands of fleet vehicles, representing a potential 16.82% reduction of all GHG emissions if all these vehicles become EVs.

Figuring out how to transition internal combustion engine (ICE) fleets to electric vehicles, as part of a company's commitment to reducing its reliance on fossil fuels, is a challenge many executives face today. With the right information on how a corporate fleet can benefit from electrification, companies can turn a traditional cost center into a revenue generator.

The Advantages of Fleet Electrification

One of the first decisions in fleet electrification is choosing between one-way (V2G) and bidirectional (V2X) charging. This decision determines whether your fleet remains a cost center or becomes a revenue generator.

Revenue Generation Through a V2X Strategy

If you choose V2G for your EV fleet, you relegate it to cost center status. On the other hand, the V2X alternative allows fleet owners to turn EVs into revenue generators. When connected to a V2X charger, an EV can recharge its batteries (V2G), send the power from its batteries to your building for cost savings (V2B), or sell it to your utility for a fee. The following graphic illustrates how V2X charging can generate positive ROI from FE, which is impossible with ICE vehicles.

Improving Asset Utilization with V2X

Non-delivery fleets can sit idle from 50-95% of the time, dramatically underutilizing these expensive assets. With few options to improve utilization, especially when vehicles, though underutilized, need to be always ready and available, an alternative is to employ a V2X strategy and turn the fleet into a revenue-producing asset as a distributed energy resource (DER).

Let's take a municipal government vehicle, for example. The organization wants to equip its building inspectors with EVs to assess their viability. During a typical year, an inspector drives 5,000 miles between 7:00 AM and 4:00 PM, leaving the vehicle parked the rest of the time. So, how does a V2X solution turn this situation around?

In a pilot supported by Fermata Energy, a municipal agency contracted with its local utility, National Grid, to sell power back during sweltering days in July. For making their EV available as a DER for 2-3 days per year, the agency earned \$4,325 annually.

The pilot achieved a 100% ROI in the first year. This enviable financial performance included all State and Federal incentives, an aggressive, limited-time \$4,000 up-front lease for two years from Nissan, one of the first EV manufacturers to approve the use of a bidirectional charger, and utility payments for energy.

Regulatory Compliance and Risk Mitigation

State and federal government agencies are issuing mandates for EV use. For example, one State has enacted legislation requiring all passenger vehicles sold in the State to be EVs by 2035. While the bill doesn't specifically target fleets, it will require organizations with passenger vehicles and light-duty trucks to accommodate EVs in the future.

Emerging Best Practices in Fleet Electrification

Fleets for administrative, customer care, health service, and other staff purposes are ideal for a V2X application. These fleets are invariably parked just as demand peaks and renewable generation power sources go offline.

When demand exceeds the available electricity supply, utilities purchase power at spot market prices, which is exceptionally expensive. And here is what makes EVs as a DER valuable to utilities and lucrative for fleet owners: these energy resources are instantly available to the host utility through a V2G arrangement. And, they are less costly, saving the utility money and generating revenue for the fleet manager.

Taking Advantage of Incentives and Other Savings

Government incentives and manufacturer programs help offset the cost difference between an EV and an ICE, albeit artificially. Depending on your location, EV chargers may also be eligible for incentives, helping reduce the cost of charging infrastructure.

Lower maintenance costs further reduce the price differential between an EV and an ICE vehicle. A 2021 GSA report estimates ICE maintenance costs at \$0.101/mile. In [a theoretical scenario with an all-EV GSA fleet](#), maintenance costs were \$0.061/mile, resulting in annual savings of \$78,000,000.

Battery Optimization

EV batteries have a "preference" for how they are charged and discharged. For example, charging an EV to 100% and letting it sit idle exposes batteries to "calendar aging," an effect more damaging than regular use. Studies find that EV batteries "like" to stay around a 60% charge level and that the practice may extend battery life.

Ensuring Vehicle Availability

Fleet vehicles must always be available to support business needs. So, a natural inclination is to use fast chargers to ensure the EV is ready to drive at any time. Fast chargers are frequently rated at 300 kW, minimizing recharge time. However, when you connect your typical 60 kW battery, you can damage and degrade it.

Today's best practice is to use a 15-kW charger to recharge the vehicle battery gently. Although these chargers take longer to reach 100%, that's rarely an issue because most fleet vehicles suitable for V2X applications rarely, if ever, discharge their batteries enough to require a lengthy recharge.

Developing an Electrification Strategy

Your ICE to EV migration strategy should address the following questions:

1. **Project ROI and Total Cost of Ownership (TCO):** A unique aspect of fleet electrification is the ability to create a revenue-generating asset instead of a cost center. Governmental incentives, rebates, and other financial inducements combined with payments from your utility need to be identified and factored into your feasibility analysis.
2. **Vehicle Maintenance:** If you maintain your ICE fleet, can these employees receive training to manage EVs? Are there local training programs paid for by government grants to retrain employees? If you outsource your maintenance, can this service provider handle EVs?
3. **Energy Costs and Revenues:** Your electricity costs will increase with an EV fleet. However, eliminating fossil fuel costs and using EVs to reduce peak power demand in your facilities helps offset the differential. And in areas that support V2X, your EVs can generate income as DERs.
4. **Risk Management:** What potential risks are associated with fleet electrification, and how can you mitigate them? Consider factors such as technology disruptions, battery life, spare parts availability, and local maintenance and repair services.

V2X is the Future of Fleet Electrification

Government agencies at all levels are seeking to decarbonize the economy through beneficial electrification. Fleet electrification presents a tempting opportunity to demonstrate early success, and organizations with fleets of light-duty cars and trucks can either be targets of government mandates or leaders, capitalizing on revenue opportunities years before government agencies require EV fleets.

Where ICE fleets are cost centers, EV fleets can become revenue generators if the company elects to use bidirectional charging in a V2X scenario. The ROI of the V2X pilots is almost unbelievable, as the

performance data demonstrates. And the upfront cost of going V2X, compared to one-way charging, may be lower.

Given the revenue potential in V2X fleet electrification, it is crucial to assemble a team to evaluate the best strategic approach to the transition. The CFO is well-positioned to provide the cost-benefit analysis of how much money a corporate fleet could earn the company rather than cost it.